# The Brent-Price Causal Effect of the 2026 Hormuz Disruption

*A Synthetic Control Approach — Milestone 3*

Cherryl Chico    Nikoloz Darsalia    Elvis Casco

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# Where we left off

**Major feedbacks:**

1. **Stronger justification that the selected donors satisfy SUTVA** — i.e. that no donor is itself contaminated by the treatment.
2. **Economic interpretation of why the models behaved as they did** during the treatment period
3. **Explore a naive counterfactual** — show what the SCM machinery buys us over simpler baselines.

Re suggested reading **Ham & Miratrix**: it concerns pre-trend identification for *matching* / *difference-in-differences* designs, not synthetic control on a price series, so its specific machinery does not transfer to our setting

# Comment 1 — how we defend donor selection (SUTVA)

*Does the asset co-move with Brent through common global factors, yet stay off the oil-supply / disruption path?*

**Approaches:**

⇒ **(Main) Economic audit.** (Abadie 2021) Screen every candidate against two contamination channels: the **oil-supply path** (energy complex, petrocurrencies — categorically out) and the **disruption itself** (Hormuz routing, Russia/Ukraine trade — screened donor-by-donor).

+ **Statistical cross-check.** Two tests of whether a donor's *own* series shifts at $T_0$ — a **permutation mean-shift** test (Chow 1960) and the **HLT** trend-break test (Harvey, Leybourne & Taylor 2009).

# Economic audit — Metals & Agriculturals

**Metals** — Silver, Platinum, Gold

- **Include:** real-rate / safe-haven / dollar channel — a monetary co-movement with Brent, off the oil-supply path.
- **Exclude:** Copper, IronOre (Russia metal supply $\sim$4–5% world), Palladium (Russia $\approx$40% world supply, spiked Mar-2022).

**Agriculturals** — Coffee, Sugar, LiveCattle

- **Include:** global trade / demand and the dollar; producers outside the conflict zones (Brazil, India, US).
- **Exclude:** Wheat, Corn (Russia+Ukraine $\approx$30% / 13% world exports), Soybeans (grain cross-elasticity), Cotton (oil$\rightarrow$polyester substitution).

# Economic audit — Equities & FX

**Equities** — SP500, Nikkei

- **Include:** global growth and the equity risk premium — the demand channel shared with oil; no Russia or Gulf-energy weight.
- **Exclude:** EM_Eq (Russia index weight); energy-heavy DAX / FTSE / TSX-60.

**FX** — AUD, JPY, CHF, CNY, INR, KRW, ZAR, MXN

- **Include:** safe-haven (JPY, CHF), EM-growth (ZAR, MXN, INR, KRW), China-demand (CNY, AUD) — dollar and growth factors.
- **Exclude:** EUR, DXY (EU energy-crisis / dollar on the invasion), GBP (UK gas); petrocurrencies (CAD, NOK, BRL, . . . ).

# Economic audit — Rates / credit & Volatility

**Rates / credit** — TLT, HYG

- **Include:** TLT (US duration) and HYG (credit-risk premium) — the discount-rate / risk side of the global factor, not oil-linked.
- **Exclude:** US10Y (Russia: inflation-expectations / flight-to-quality loads it onto the treatment).

**Volatility** — VIX

- **Include:** the single risk-on / risk-off switch — the fear premium that also lifts oil in stress.
- **Exclude:** none — one volatility proxy, kept deliberately to avoid over-weighting the risk channel.

**Net of the audit: 19 of 32 candidates retained** (shared pool, both events — 13 dropped for Russia exposure).

# Statistical cross-check — shifts at $T_0$

- **Permutation mean-shift** (Chow 1960): split the donor's own series at $T_0$, take the mean gap (after − before), and compare it to thousands of random before/after relabelings — an unusually large gap signals a real shift. Nonparametric, no distributional assumption.
- **HLT trend-break** (Harvey, Leybourne & Taylor 2009): tests whether the donor's own *trend* breaks at $T_0$, valid whether the price is stationary or a random walk. Brent-free, so a break implicates the donor itself, not Brent's own $T_0$ move.

| Flagged at $T_0$ (of 19 retained) | Hormuz 2026 | Russia 2022 |
|---|---|---|
| HLT trend-break | 0 / 19 | 0 / 19 |
| Permutation mean-shift | 0 / 19 | 1 / 19 (CNY) |

HLT flags nothing. The lone flag — **CNY** — is the concurrent China zero-COVID reopening, not Russia-treatment, and is kept on the audit's judgment. *Tests inform; the audit decides.*

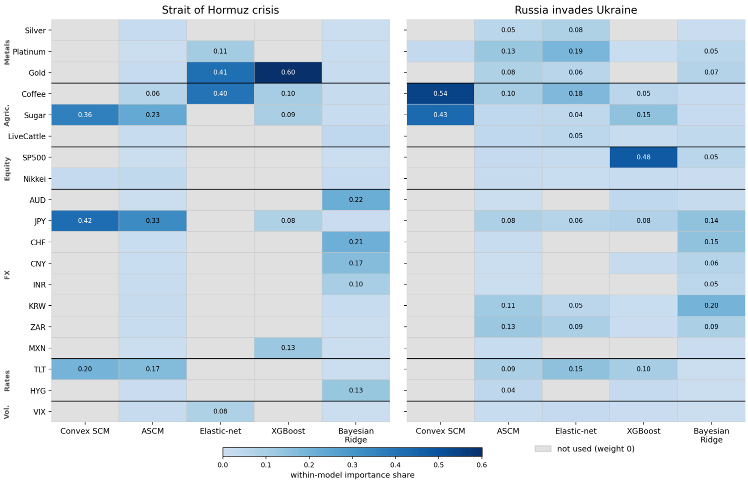
# Comment 2 — why the models behaved as they did

*Why did the five models move (and disagree) the way they did during the event — and does that behaviour inflate the premium?*

**Approaches:**

1. **Behaviour via donor weights.** Which donors each model leans on — explains the gap (donors miss the oil-specific shock) and the Russia divergence (different donors).
2. **Leave-one-out robustness.** Does any single donor drive the gap?
3. **Signed contamination bias.** Are those donors themselves moved by the treatment (demand destruction)? Decompose each gap into a clean part and a donor-contamination part, and *sign* it.

# Behaviour via donor weights




**Hormuz.** Weight on safe-haven / non-oil donors (Gold, JPY, Sugar) — they track common risk but *not* the chokepoint shock ⇒ synthetic undershoots Brent ⇒ positive gap.

**Russia — divergence is donor-driven.** Models lean on *different* donors (Coffee/Platinum vs SP500); Bayesian Ridge degenerates ($w \approx 1.8$–$2.4$) ⇒ dropped from the IQR.

Each column = that model's own importance metric (convex weight / $|\hat{\beta}_j|$ / XGBoost gain), normalised over the 19 donors. Gray = not used.

# Leave-one-out robustness

**Procedure.** Drop one influential donor ($w > 5\%$) at a time, refit, recompute the ATT. Report the **range** (max − min, pp) across the leave-one-donor-out refits — a small range means no single donor drives the result.

| **Model** | **Hormuz LOO range (pp)** | **Russia LOO range (pp)** |
|---|---:|---:|
| Convex SCM | 8.2 | 26.8 |
| ASCM | **5.7** | 19.1 |
| Elastic-net | 6.7 | 18.8 |
| XGBoost | 6.1 | **13.8** |
| Bayesian Ridge | 10.4 | 27.3 † |

Source: data/validation/inference_loo_{event}_{model}.csv.

- **Hormuz: tight** (∼6–10 pp on a ∼59% effect) — no single donor flips the result; ASCM tightest.
- **Russia: wide** (14–27 pp). Bayesian Ridge († range crosses zero, −21.9 to +5.4) is LOO-fragile, consistent with its degenerate fit; XGBoost is the most stable. Reinforces Russia as a magnitude check, not the headline.

# Signed contamination bias

**The critique.** If high oil *crowds out demand*, activity-proxy donors are pushed *down*, dragging the synthetic down → contamination in the gap. **Which way, and how big?** We sign each linear model's net donor-bias ($-\sum_j w_j\,\delta_j$) over the full post-window:

| Model | Hormuz net bias | Russia net bias |
|---|---|---|
| Convex SCM | −1.3% (conservative) | +21.4% |
| ASCM | −2.1% (conservative) | +12.7% |
| Elastic-net | −2.4% (conservative) | +20.6% |
| Bayesian Ridge | −7.6% (conservative) | −7.1% |

Linear models only (XGBoost nonlinear). Source: contamination_bias_decomposition.csv.

- **Hormuz: bias negative** for all linear models → demand-destruction makes the SCM *under*-state the premium, so the headline **∼59%** is a **lower bound**.
- **Russia: bias positive** → retained cyclical donors (Coffee, Platinum, SP500) were pushed *down*, dragging the synthetic down and *inflating* the gap → Russia is a magnitude check, not the headline.

# Comment 3 — what does the SCM buy us over naive baselines?

> *If you ignore the donor pool and just extrapolate Brent's own past, what counterfactual do you draw — and how far off is the premium?*

**Appraoches:**

1. **Donor-free baselines.** Build univariate counterfactuals on Brent alone — random walk (flat / drift), linear pre-trend, ARIMA and variants
2. **Benchmark the premium.** Project each over the treatment window and compare its implied gap to the donor-informed SCM — exposing baselines that freeze or extrapolate blindly.

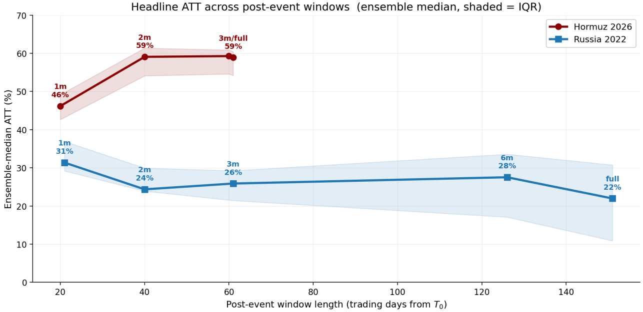
# Donor-free naive baselines

**What does the donor-informed SCM buy us over baselines that ignore the donors?**


- **Freeze / revert baselines** (flat & AR random walks, mean-reversion) *understate* — Hormuz ∼52%, Russia ∼9%.
- **Trend extrapolation** overshoots — linear pre-trend gives Hormuz **73.8%** (overstate); on Russia the extrapolated rally *flips the gap negative* (linear −4.2%, RW+drift −6.9%).
- **SCM** alone tracks the donor-implied path: **58.9%** Hormuz, **22.0%** Russia.

# Summary — headline ATT across post-event windows

1. **SUTVA** — audit (oil-supply / disruption path), corroborated by the breakpoint test.
2. **Model behaviour** — weight on risk donors explains the gap; contamination bias signed (Hormuz conservative); robust to leave-one-out.
3. **Naive baselines** — random walks understate, trend; the SCM sits between.




**Hormuz** builds to $\sim$**59%** by month 2–3 and holds (IQR $[54, 60]$); **Russia** peaks $\sim$31% (1m) and settles $\sim$22% (full) — the magnitude check.

# Thank you

Questions and discussion

# Appendix — in-space placebo (treated-unit inference)

**Procedure** (Abadie 2010): refit the SCM treating each donor as treated; rank Brent's post/pre RMSPE ratio against the $N$=20 units (Brent + 19 donors). **Permutation** $p$ = **rank** / $N$; floor $p = 1/20 = 0.05 \rightarrow$ rank 1 is the strongest signal the 19-donor pool can produce.

| Model | Hormuz 2026 rank $\rightarrow p$ | Russia 2022 rank $\rightarrow p$ |
|---|---|---|
| Convex SCM | **1 → 0.05** | 10 → 0.50 |
| ASCM | **1 → 0.05** | 12 → 0.60 |
| Elastic-net | **1 → 0.05** | 8 → 0.40 |
| XGBoost | **1 → 0.05** | 4 → 0.20 |
| Bayesian Ridge | **1 → 0.05** | 12 → 0.60 |

Source: data/validation/inference_inspace_brent_summary.csv (19-donor shared pool).

- **Hormuz:** Brent ranks 1 for **all five models** → its break beats every placebo; strongest signal the pool allows.
- **Russia:** mid-pack (rank 8–12) for the linear models; XGBoost is best at rank 4 ($p$=0.20) but still not significant — consistent with Russia as a magnitude check, not a clean inferential case.

# Appendix — walk-forward CV RMSE (model fit)

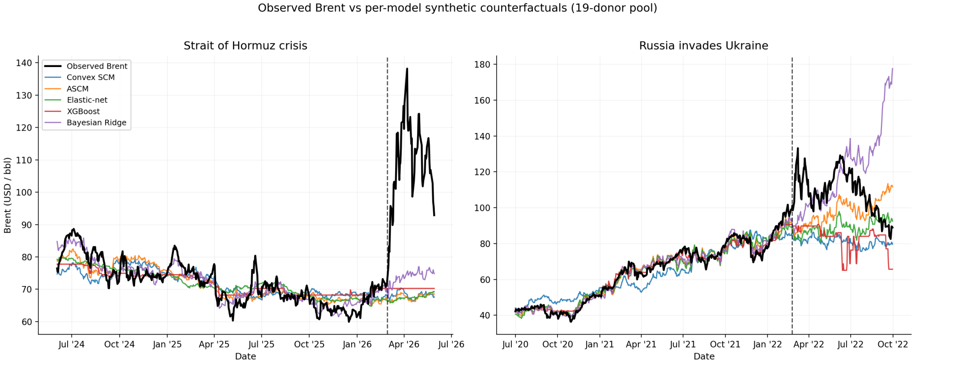
5-fold expanding-window walk-forward CV (§5a): **val RMSE** = pooled out-of-sample error, **train RMSE** = cross-fold mean. **val/train** > 2 is a heuristic over-fit flag (flagged models are kept — the ensemble takes the median).

| Event | Model | train RMSE | val RMSE | val/train | flag |
|---|---|---|---|---|---|
| Russia | Convex SCM | 0.113 | 0.141 | 1.25 | OK |
| Russia | ASCM | 0.046 | 0.095 | **2.06** | overfit |
| Russia | Elastic-net | 0.051 | 0.102 | 1.99 | OK |
| Russia | XGBoost | 0.039 | 0.130 | **3.31** | overfit |
| Russia | Bayesian Ridge | 0.034 | 0.105 | **3.09** | overfit |
| Hormuz | Convex SCM | 0.060 | 0.083 | 1.40 | OK |
| Hormuz | ASCM | 0.049 | 0.075 | 1.54 | OK |
| Hormuz | Elastic-net | 0.052 | 0.079 | 1.51 | OK |
| Hormuz | XGBoost | 0.045 | 0.083 | 1.85 | OK |
| Hormuz | Bayesian Ridge | 0.034 | 0.078 | **2.28** | overfit |

RMSE on log-Brent. Source: data/validation/validation_summary.csv. **Russia** 2/5 clean (volatile 2020–22 pre-period); **Hormuz** 4/5 clean — flagged models stay in, the ensemble median is robust.

# Appendix — per-model synthetic paths

Observed Brent (black) vs each model's synthetic counterfactual. Pre-$T_0$ the synthetics *track* Brent (the fit); post-$T_0$ they diverge — the gap between observed and synthetic is the premium.




Convex SCM & ASCM are the headline pair; Bayesian Ridge's Russia path blows up (degenerate weights, $w$ outside the simplex) — which is why it is down-weighted in the ensemble.